



Student Name: _____

Maths class: _____

James Ruse Agricultural High School

2023

YEAR 12 Task 1 HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 75 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 6–10, show relevant mathematical reasoning and/ or calculations

Total marks: 45

Section I – 5 marks

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 40 marks

- Attempt Questions 6–10
- Allow about 68 minutes for this section

Start your answer to each question on a separate piece of paper.

Section I

5 marks

Attempt Questions 1-5

Allow about 7 for this section

Write your answers on the multiple choice answer sheet provided.

1. Evaluate $i^{2022} + i^{2023} + i^{2024}$

- (A) i
- (B) $-i$
- (C) 1
- (D) -1

2. A complex number z is defined such that $\operatorname{Re}(z) > 0$ and $\operatorname{Im}(z) < 0$. Which of the following is a suitable range for $\arg(z^3)$?

- (A) $\left(\frac{\pi}{2}, 2\pi\right)$
- (B) $\left(0, \frac{\pi}{2}\right)$
- (C) $(0, 2\pi)$
- (D) $\left(-\frac{\pi}{2}, 0\right)$

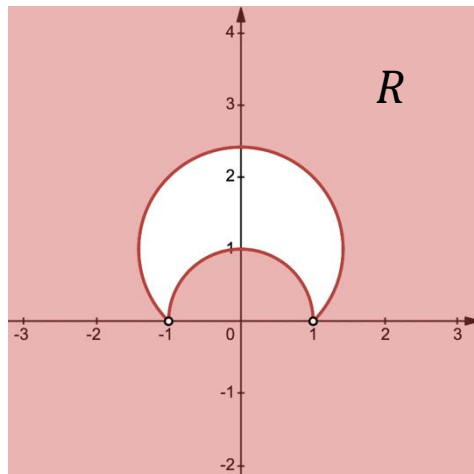
3. If the complex numbers ω and z represent two diagonally opposite vertices of a square, then the other two vertices could be given by the complex numbers:

- (A) $\frac{1}{2}(z - w) + \frac{1}{2}i(z - w)$ and $\frac{1}{2}(z - w) - \frac{1}{2}i(w - z)$
- (B) $\frac{1}{2}(\omega + z) + \frac{1}{2}i(\omega + z)$ and $\frac{1}{2}(\omega + z) - \frac{1}{2}i(\omega + z)$
- (C) $\frac{1}{2}(\omega - z) + \frac{1}{2}i(\omega - z)$ and $\frac{1}{2}(\omega - z) - \frac{1}{2}i(\omega - z)$
- (D) $\frac{1}{2}(\omega + z) + \frac{1}{2}i(\omega - z)$ and $\frac{1}{2}(\omega + z) - \frac{1}{2}i(\omega - z)$

4. Let $a, b, c \in \mathbb{R}$. If the equation $x^4 + ax^3 + bx^2 + cx + 1 = 0$ has no real roots and if at least one root is of modulus one, then which of the following is true?

- (A) $b = c$
- (B) $a = b$
- (C) $a = c$
- (D) None of the above

5. Which of the following options represents the shaded region, R , below?



- (A) $\frac{\pi}{2} \leq \text{Arg} \left(\frac{z-1}{z+1} \right) \leq \frac{3\pi}{4}$
- (B) $\frac{\pi}{4} \leq \text{Arg} \left(\frac{z-1}{z+1} \right) \leq \frac{\pi}{2}$
- (C) $\text{Arg} \left(\frac{z-1}{z+1} \right) \leq \frac{\pi}{2}$ and $\text{Arg} \left(\frac{z-1}{z+1} \right) \geq \frac{3\pi}{4}$
- (D) $\text{Arg} \left(\frac{z-1}{z+1} \right) \leq \frac{\pi}{4}$ and $\text{Arg} \left(\frac{z-1}{z+1} \right) \geq \frac{\pi}{2}$

Section II begins on the next page

Section II

40 marks

Attempt Questions 6-10

Allow about 68 minutes for this section

Start each question on a new page. Extra paper is available.

In Questions 6-10, your responses should include relevant mathematical reasoning and/ or calculations.

Question 6 (8 marks) Start a new page.

- (a) If $z = 3 + 4i$ and $\omega = 2 - i$, simplify the following, giving your answer in the form $x + iy$

(i) $z - 2\omega$

1

(ii) $z\bar{z} - \omega\bar{\omega}$

1

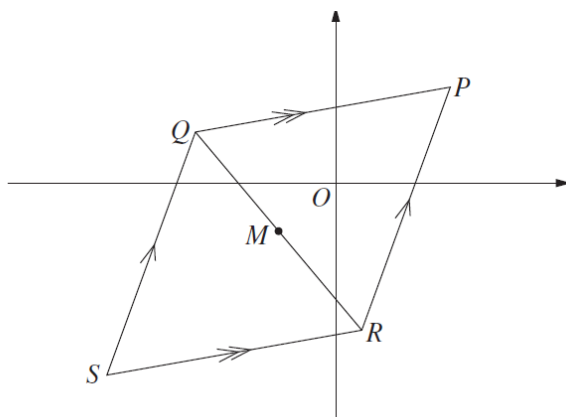
(iii) $\frac{z}{\omega}$

2

- (b) Given two complex numbers $z = \sqrt{3} + i$ and $\omega = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$. Express $\frac{z\omega}{4i}$ in the form $r(\cos\theta + i\sin\theta)$.

2

- (c) The point P on the Argand diagram represents the complex number z . The points Q and R represent the points ωz and $\bar{\omega}z$ respectively, where $\omega = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$. The point M is the midpoint of QR .



- (i) Find the complex number representing M in terms of z .

1

- (ii) The point S is chosen so that $PQSR$ is a parallelogram. Find the complex number represented by S .

1

Question 7 (8 marks) Start a new page.

- (a) The complex number z is such that $|z| = 1$ and $\arg(z) = \frac{\pi}{4}$.
- (i) Find the coordinates of points A representing z^2 , and B representing $z^2 - \bar{z}$ respectively and plot the points on a common half-page Argand diagram. 2
- (ii) Prove that $|z^2 - \bar{z}| = |z + 1|$. 2
- (b) (i) Find the square roots of $-63 + 16i$ 2
- (ii) Hence or otherwise, solve for z in the equation $2z^2 + (\sqrt{63})iz - 2i = 0$. Write your answers in the form $x + iy$. 2

Question 8 (8 marks) Start a new page.

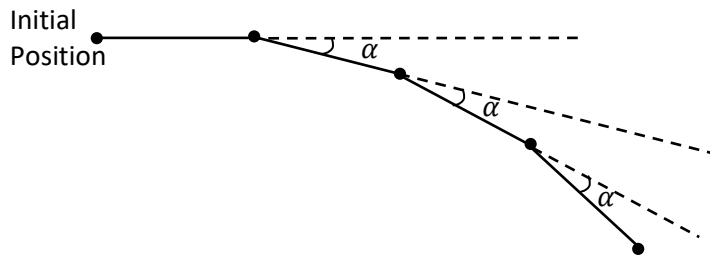
- (a) (i) Show that $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ 1
- (ii) Hence or otherwise, prove that the solution to θ in the equation $\cos \theta = 2$ is that of $e^{i\theta} = 2 \pm \sqrt{3}$ 1
- (b) The complex number z satisfies the relationship $5(z + i)^n = (4 + 3i)(1 + iz)^n, n \in \mathbb{R}$. Show that z is a real number. 3
- (c) Find all complex numbers z satisfying both $\text{Arg}(z^2 - 4) = \frac{5\pi}{6}$ and $\text{Arg}(z^2 + 4) = \frac{\pi}{3}$ 3

Question 9 (8 marks) Start a new page.

- (a) (i) Let $a \in \mathbb{C}$ and $b \in \mathbb{R}, b \geq 0$. Show that the equation 2
- $$z\bar{z} = a\bar{z} + \bar{a}z + b$$
- for $z \in \mathbb{C}$ can be written in the form $|z - a| = r$, where $r \geq 0$.
- Let A and B be two points on the complex plan representing $2 + 3i$ and $1 + 2i$ respectively. The point P , representing the complex number z , moves such that $PA = \sqrt{2}PB$.
- (ii) Show that the equation of the path traced out by P is a circle with equation 3
- $$\mathcal{C}_1: z\bar{z} = i\bar{z} - iz + 3.$$
- State its centre and radius.
- Let Q , representing ω , be a point on the circle $\mathcal{C}_1$.
- (iii) Show that the circle $\mathcal{C}_2$ given by $\left|z - \left(\omega + \frac{\omega - i}{|\omega - i|}\right)\right| = 1$ touches $\mathcal{C}_1$ at Q externally. 3

Question 10 (8 marks) Start a new page.

- (a) A student is in the initial stages of programming a simple robotic car. So far, the car only moves in a straight line 1 unit at a time. After each 1 unit move forward, it changes direction by turning to the right through an acute angle of α radians and then moving forward another 1 unit. An example of a possible path is shown below with 4 forward moves.



During a test run, the car starts from the origin and initially moves to the right. It moves forward a total of 24 times before finally stopping.

- (i) Explain why the final position of the car can be represented by the expression **1**

$$1 + e^{-i\alpha} + e^{-i2\alpha} + \dots + e^{-i23\alpha}.$$

- (ii) If the car's final position is the same as its initial position, find all possible values of α . **2**

- (iii) If the **only** time the car reaches its initial position during its path is when it arrives at its final position, find all possible values of α , giving reasons. **2**

The test run is considered to be *successful* if the car stops within a one unit distance from its starting point.

- (iv) Show that for the test to be considered *successful*, then α must satisfy $|\sin(12\alpha)| \leq \left| \sin\left(\frac{\alpha}{2}\right) \right|$. **3**

END OF EXAMINATION

2023 Year 12 Task 1 Extension 2

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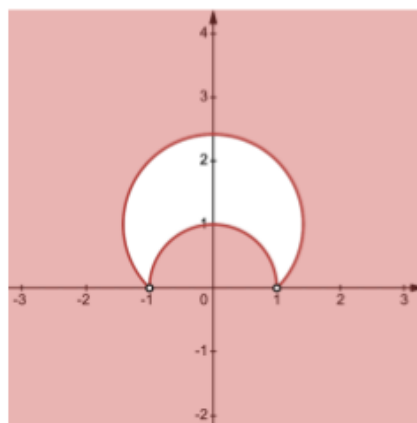
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 (D) $\operatorname{Arg}\left(\frac{z-1}{z+1}\right) \leq \frac{\pi}{4}$ and $\operatorname{Arg}\left(\frac{z-1}{z+1}\right) \geq \frac{\pi}{2}$

Question 6

(a) $Z = 3 + 4i$, $w = 2 - i$

(i) $Z - 2w = 3 + 4i - 2(2 - i)$

$$= 3 + 4i - 4 + 2i$$

$$= -1 + 6i \quad \checkmark$$

(ii) $Z\bar{Z} - w\bar{w} = |Z|^2 - |w|^2$

$$= 3^2 + 4^2 - (2^2 + (-1)^2)$$

$$= 9 + 16 - 4 - 1$$

$$= 20 \quad \checkmark$$

(iii) $\frac{Z}{w} = \frac{Z\bar{w}}{|w|^2} = \frac{(3+4i)(2+i)}{2^2+(-1)^2} = \frac{6+3i+8i-4}{5} = \frac{2+11i}{5} \quad \checkmark$

(b) $Z = \sqrt{3} + i$ and $w = 6\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$

$$= 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) \quad \checkmark$$

$$\frac{Zw}{4i}$$

$$= \frac{2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) \times 6\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)}{4i}$$

$$= 3\left(\cos \left(\frac{\pi}{6} + \frac{2\pi}{3} - \frac{\pi}{2}\right) + i \sin \left(\frac{\pi}{6} + \frac{2\pi}{3} - \frac{\pi}{2}\right)\right)$$

$$= 3\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right) \quad \checkmark$$



(c)

(i) The complex number representing M

$$\begin{aligned} \text{is } \frac{\omega Z + \bar{\omega} Z}{2} &= \frac{(\omega + \bar{\omega}) Z}{2} \\ &= \frac{2 \cos 2\pi/3}{2} Z \\ &= \cos \frac{2\pi}{3} Z \\ &= -\frac{Z}{2} \quad \checkmark \end{aligned}$$

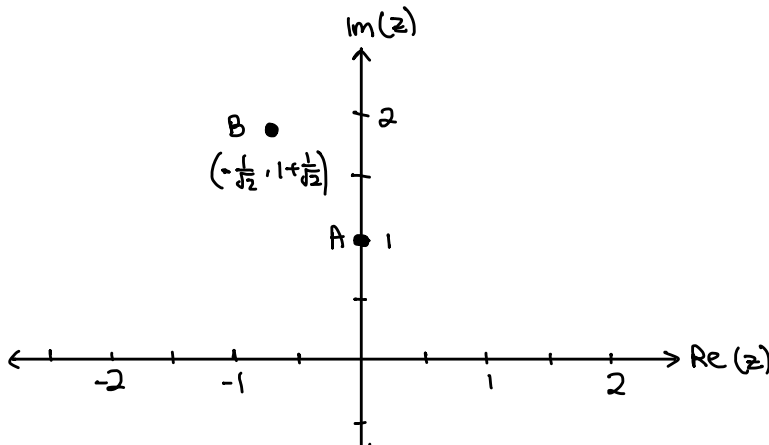
(ii) SQ = RP

Let α the complex number representing S

$$\omega Z - \alpha = Z - \bar{\omega} Z$$

$$\begin{aligned} \alpha &= (\omega + \bar{\omega}) Z - Z \\ &= -Z - Z \\ &= -2Z \quad \checkmark \end{aligned}$$

MATHEMATICS Extension 2: Question...7....

Suggested Solutions	Marks	Marker's Comments
a) i) $z^2 = \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)^2$ $= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$ (De Moivre's) $= i$ $z^2 - \bar{z} = i - (\cos \frac{\pi}{4} - i \sin \frac{\pi}{4})$ $= i - \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$ $= -\frac{1}{\sqrt{2}} + i \left(1 + \frac{1}{\sqrt{2}}\right)$  ii) $z^2 - \bar{z} = z + 1$ LHS = $\left -\frac{1}{\sqrt{2}} + i \left(1 + \frac{1}{\sqrt{2}}\right) \right$ $= \sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2 + \left(1 + \frac{1}{\sqrt{2}}\right)^2}$ $= \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(1 + \frac{1}{\sqrt{2}}\right)^2}$ RHS = $\left \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i + 1 \right$ $= \sqrt{\left(\frac{1}{\sqrt{2}} + 1\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$ $= \text{LHS}$ $\therefore z^2 - \bar{z} = z + 1$	1 1 1 1	for values of z^2 and $z^2 - \bar{z}$ for plotting on Argand diagram must show the $\left(-\frac{1}{\sqrt{2}}\right)^2$

MATHEMATICS Extension 2: Question...7...

Suggested Solutions	Marks	Marker's Comments
b) i) $\sqrt{-63+16i} = \sqrt{1+2(8i)+64i^2}$ $= \sqrt{1+2(8i)+(8i)^2}$ $= \sqrt{(1+8i)^2}$ $= \pm (1+8i)$ ii) $z = \frac{-\sqrt{63}i \pm \sqrt{(\sqrt{63}i)^2 - 4(2)(-2i)}}{2(2)}$ $= \frac{-3\sqrt{7}i \pm \sqrt{-63+16i}}{4}$ $= \frac{-3\sqrt{7}i \pm (1+8i)}{4}$ $= \frac{1}{4} + \frac{8-3\sqrt{7}}{4}i \text{ or } -\frac{1}{4} - \frac{8+3\sqrt{7}}{4}i$	1 1 1 1	

MATHEMATICS EXT 2: Question 8

Suggested Solutions	Marks	Marker's Comments
ai, $e^{i\theta} = \cos\theta + i\sin\theta$ ① $e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$ $e^{-i\theta} = \cos\theta - i\sin\theta$ ② ① + ②: $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ $\therefore \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$	①	
ii, $\cos\theta = 2$ $\frac{e^{i\theta} + e^{-i\theta}}{2} = 2$ $e^{i\theta} + e^{-i\theta} = 4$ $e^{i\theta} - 4 + e^{-i\theta} = 0$ $(e^{i\theta})^2 - 4e^{i\theta} + 1 = 0$ $e^{i\theta} = \frac{4 \pm \sqrt{16 - 4 \times 2 \times 1}}{2}$ $= \frac{4 \pm \sqrt{12}}{2}$ $e^{i\theta} = 2 \pm \sqrt{3}$	①	
b, $5(z+i)^{\wedge} = (4+3i)(1+iz)^{\wedge}$ $5(z+i)^{\wedge} = (4+3i)(1+iz)^{\wedge}$ ① $5 z+i ^{\wedge} = 5 1+iz ^{\wedge}$ $z+i ^{\wedge} = 1+iz ^{\wedge}$ $z+i = 1+iz$ $z+i = i z-i$ $z+i = z-i$ ①	①	Recognizing to take modulus of both sides Use modulus properties to simplify expression
$\therefore z$ lies on the perpendicular bisector of i and $-i$, which is the real axis, i.e. z must be a real number.	①	Valid explanation

MATHEMATICS EXT 2: Question 8

Suggested Solutions

Marks

Marker's Comments

c, ALGEBRAIC APPROACH:

$$\text{Let } z^2 = x+iy$$

$$\therefore \text{Arg}(x+iy-4) = \frac{5\pi}{6}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x-4}\right) = \frac{5\pi}{6}$$

$$\Rightarrow \frac{y}{x-4} = -\frac{1}{\sqrt{3}}$$

$$\Rightarrow y\sqrt{3} = 4-x$$

①

$$\therefore \text{Arg}(x+iy+4) = \frac{\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x+4}\right) = \frac{\pi}{3}$$

$$\Rightarrow \frac{y}{x+4} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow y = \sqrt{3}x + 4\sqrt{3}$$

②

Sub ② into ①:

$$\sqrt{3}(\sqrt{3}x + 4\sqrt{3}) = 4-x$$

$$3x + 12 = 4-x$$

$$4x = -8$$

$$x = -2$$

$$\therefore y = 2\sqrt{3}$$

$$\therefore z^2 = -2 + 2\sqrt{3}i$$

$$= 1 + 2 \times 1 \times \sqrt{3}i + (\sqrt{3}i)^2$$

$$z^2 = (1 + \sqrt{3}i)^2$$

$$\therefore z = \pm(1 + \sqrt{3}i)$$

①

①

①

MATHEMATICS EXT 2: Question 8		
Suggested Solutions	Marks	Marker's Comments
<u>Note:</u> Letting $z = x + iy$ can work, but it is very ugly. Since it COULD work, one mark was assigned for getting to $\frac{2xy}{x^2 - y^2 - 4} = -\frac{1}{\sqrt{3}} \quad \& \quad \frac{2xy}{x^2 - y^2 + 4} = \sqrt{3}$		

MATHEMATICS EXT 2: Question 8		
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MATHEMATICS EXT 2: Question 8

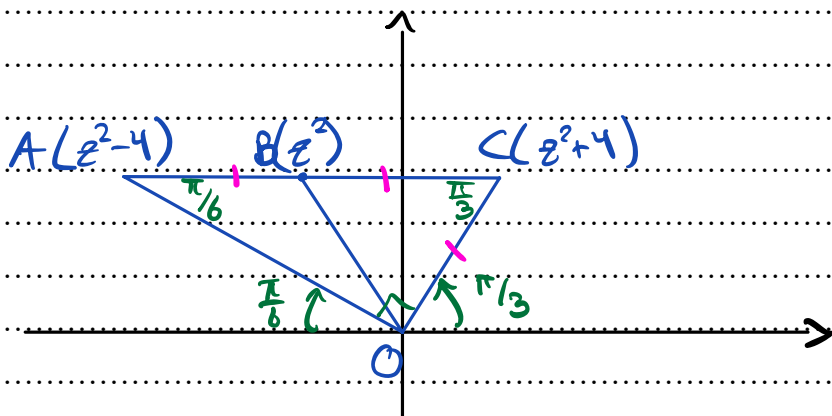
Suggested Solutions

Marks

Marker's Comments

GEOMETRIC METHOD:

$$\operatorname{Arg}(z^2 - 4) = \frac{5\pi}{6} \quad \& \quad \operatorname{Arg}(z^2 + 4) = \frac{\pi}{3}$$



$$\begin{aligned} \angle OCB &= \frac{\pi}{3} & (\text{alternate angles}) \\ \angle OAC &= \frac{\pi}{6} & (\text{alternate angles}) \end{aligned}$$

$$\text{Now } \frac{OC}{AC} = \sin \frac{\pi}{6}$$

$$\frac{OC}{AC} = \frac{1}{2}$$

$$\begin{aligned} \therefore OC &= \frac{1}{2} AC \\ &= 4 \end{aligned}$$

$$\text{But } BC = 4 = OC$$

$\therefore \triangle OBC$ is isosceles with included angle of $\frac{\pi}{3}$

$\therefore \triangle OBC$ is equilateral

$$\therefore OB = 4$$

$$\Rightarrow z^2 = 4 e^{\frac{2\pi i}{3}}$$

$$\begin{aligned} \Rightarrow z &= \pm 2 e^{\frac{\pi i}{3}} \\ &= \pm (1 + \sqrt{3}i) \end{aligned}$$

Question 9

a)

i. $z\bar{z} = a\bar{z} + \bar{a}z + b$

Let $z = x + iy$ and $a = c + id$

$$\begin{aligned}x^2 + y^2 &= (c + id)(x - iy) + (c - id)(x + iy) + b \\&= (cx - cyi + dxi + dy) + (cx + cyi - dxi + dy) + b \\&= 2(cx + dy) + b\end{aligned}$$

1 mark

$$x^2 - 2cx + y^2 - 2dy = b$$

$$(x^2 - 2cx + c^2) + (y^2 - 2dy + d^2) = b + c^2 + d^2$$

$$(x - c)^2 + (y - d)^2 = b + c^2 + d^2$$

$$|(x - c) + (y - d)i|^2 = b + c^2 + d^2$$

$$|z - a|^2 = b + c^2 + d^2$$

$$\begin{aligned}|z - a| &= \sqrt{b + |a|^2} \\&= r \quad (\text{Since } b, c, d \in \mathbb{R})\end{aligned}$$

$$|z - a| = r \text{ where } r = \sqrt{b + |a|^2}$$

1 mark

ii. Let $P(x, y)$ represent the complex number $x + iy$

$$PA = \sqrt{2}PB \Rightarrow |PA|^2 = 2|PB|^2$$

$$(x - 2)^2 + (y - 3)^2 = 2(x - 1)^2 + 2(y - 2)^2 \quad \text{1 mark for interpreting condition algebraically}$$

(Note: It was given $PA = \sqrt{2}PB$, NOT $\overrightarrow{PA} = \sqrt{2}\overrightarrow{PB}$, if you equate the two without a square root or having squared both sides you don't receive the first mark)

$$(x^2 - 4x + 4) + (y^2 - 6y + 9) = (2x^2 - 4x + 2) + (2y^2 - 8y + 8)$$

$$x^2 + y^2 - 2y = 3$$

$$x^2 + (y^2 - 2y + 1) = 4$$

$$x^2 + (y - 1)^2 = 4$$

$\therefore$ The vector equation of this circle is given by $|z - i| = 2$

which can be expressed as $z\bar{z} = i\bar{z} - iz + b$

1 mark

Since $r = \sqrt{b + |a|^2}$ (proven in part i)

$$2 = \sqrt{b + 1^2}$$

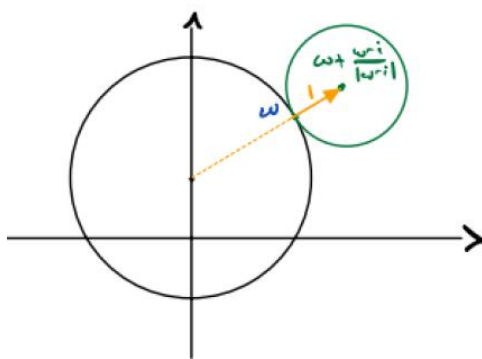
$$b + 1 = 4$$

$$b = 3$$

$\therefore P$ traces out a path given by $z\bar{z} = i\bar{z} - iz + 3$ which is a circle with radius 2 and centre (0,1).

1 mark

- iii. The circles are externally tangential if the distance between the centres are equal to the sum of the two radii. That is,



1 mark for interpreting what is required geometrically

$$C_2 \text{ is given by: } \left| z - \left(\omega + \frac{\omega - i}{|\omega - i|} \right) \right| = 1$$

$$\omega \text{ lies on } C_1 \Rightarrow |\omega - i| = 2$$

$$\therefore \left| z - \left(\omega + \frac{\omega - i}{2} \right) \right| = 1 \quad \text{1 mark for simplifying equation of } C_2 \text{ using the fact that } \omega \text{ lies on } C_1$$

$$\left| z - \left(\frac{3\omega - i}{2} \right) \right| = 1$$

$\therefore$ Distance between the two centres is given by:

$$\left| i - \left(\frac{3\omega - i}{2} \right) \right| = \left| \frac{2i - 3\omega + i}{2} \right|$$

$$= \left| \frac{3i - 3\omega}{2} \right|$$

$$= \frac{3}{2} |i - \omega|$$

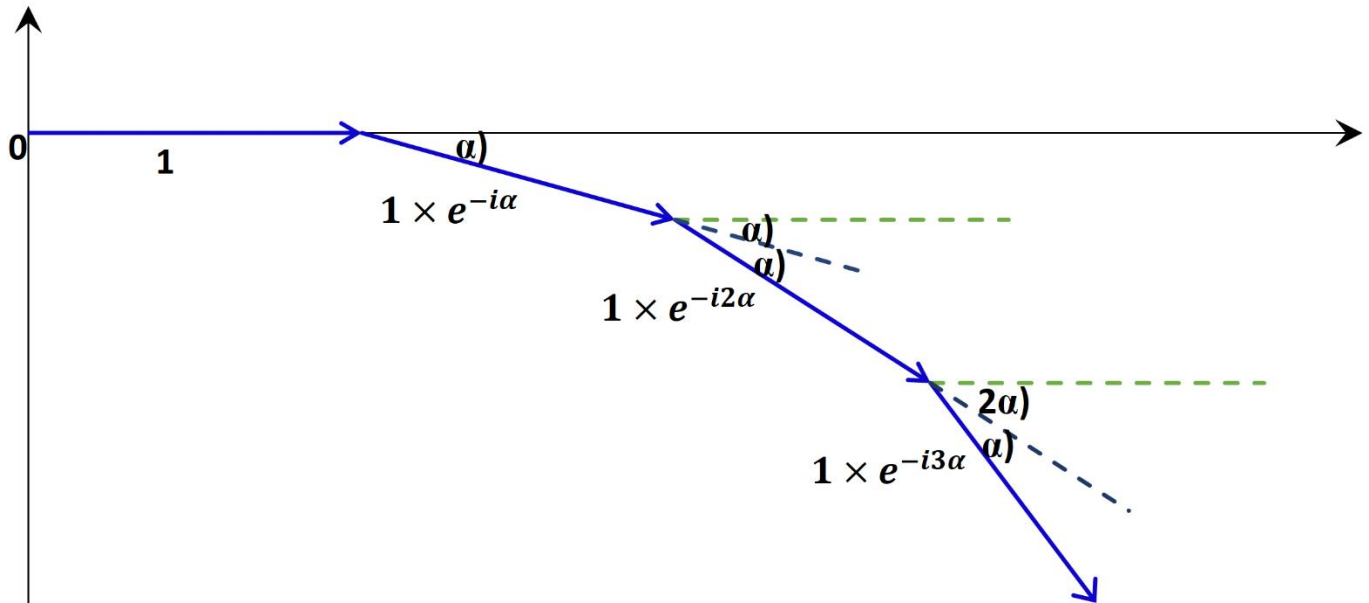
$$= \frac{3}{2} \times 2$$

$$= 3$$

1 mark for completing the proof

Question 10

- i. The path of the car is shown below.



Each forward movement has a distance of 1 unit and turns to the right by α each time. This is equivalent to multiply by $e^{-i\alpha}$ each time, where $0 < \alpha < \frac{\pi}{2}$.

Therefore, the final position is given by :

$$1 + 1 \times e^{-i\alpha} + 1 \times (e^{-i\alpha})^2 + 1 \times (e^{-i\alpha})^3 + \dots 1 \times (e^{-i\alpha})^{23} \\ = 1 + e^{-i\alpha} + e^{-i2\alpha} + e^{-i3\alpha} + \dots + e^{-i23\alpha}$$

(1 mark for providing correct explanation: stating each movement has a distance of 1 unit, and turns to the right by α is to multiply by $e^{-i\alpha}$)

- ii. The car's final position is the same as its initial position, which means:

$$1 + e^{-i\alpha} + e^{-i2\alpha} + e^{-i3\alpha} + \dots + e^{-i23\alpha} = 0$$

That is:

$$\frac{1 - e^{-24i\alpha}}{1 - e^{-i\alpha}} = 0 \quad (1 \text{ mark})$$

$$e^{-24i\alpha} = 1,$$

(The denominator can not be zero since $0 < \alpha < \frac{\pi}{2}$)

Therefore:

$$24\alpha = 2k\pi$$

$$\alpha = \frac{2k\pi}{24}$$

Where:

$$0 < \frac{2k\pi}{24} < \frac{\pi}{2}, \quad k \in \mathbb{Z}$$

$$0 < k < 6$$

$$\therefore k = 1, 2, 3, 4, 5$$

Therefore, possible values of α are

$$\frac{\pi}{12}, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{10\pi}{24} \quad (1 \text{ mark})$$

- iii. In order for the car to reach the original position for the first time, we need values of k which has no factors in common with 24 except 1.

That is $k = 1, 5$

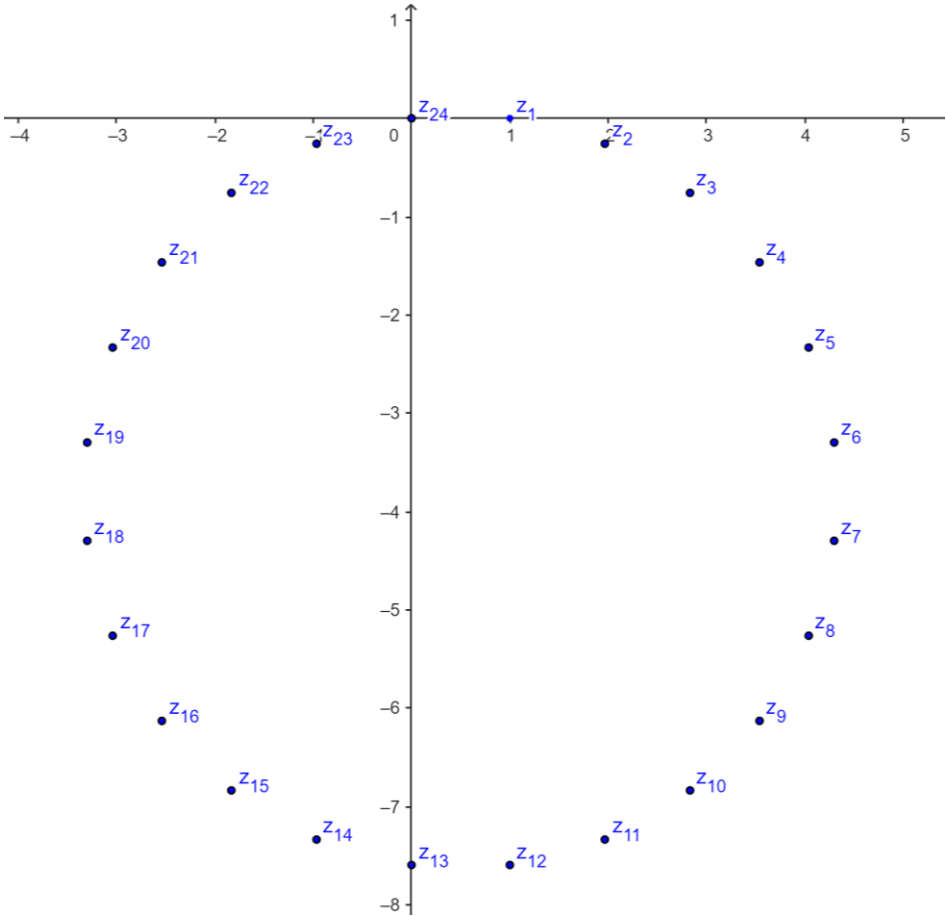
$$\therefore k = 1, \quad \alpha = \frac{\pi}{12}$$

$$k = 5, \quad \alpha = \frac{10\pi}{24}$$

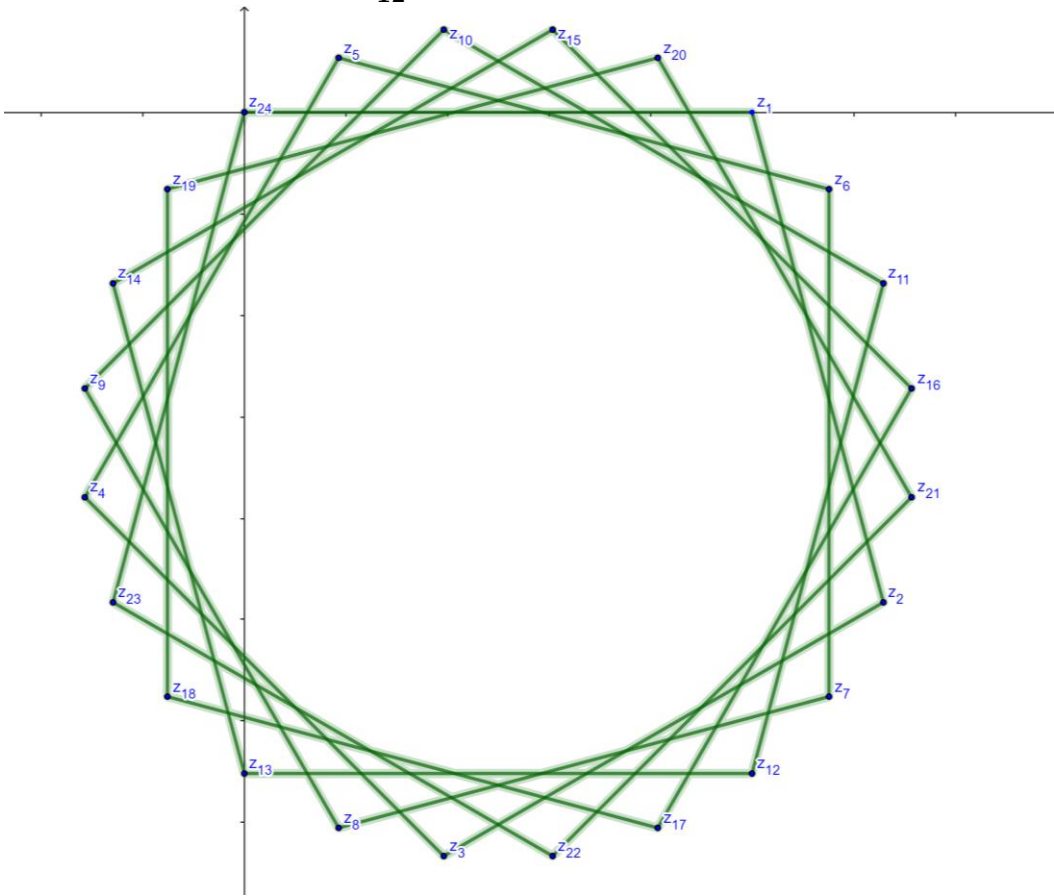
(1 mark for providing correct reason, 1 mark for correct answer)

(Common mistakes: many students assume that the path of the car must be regular polygons. The paths for three cases are shown below for your reference.)

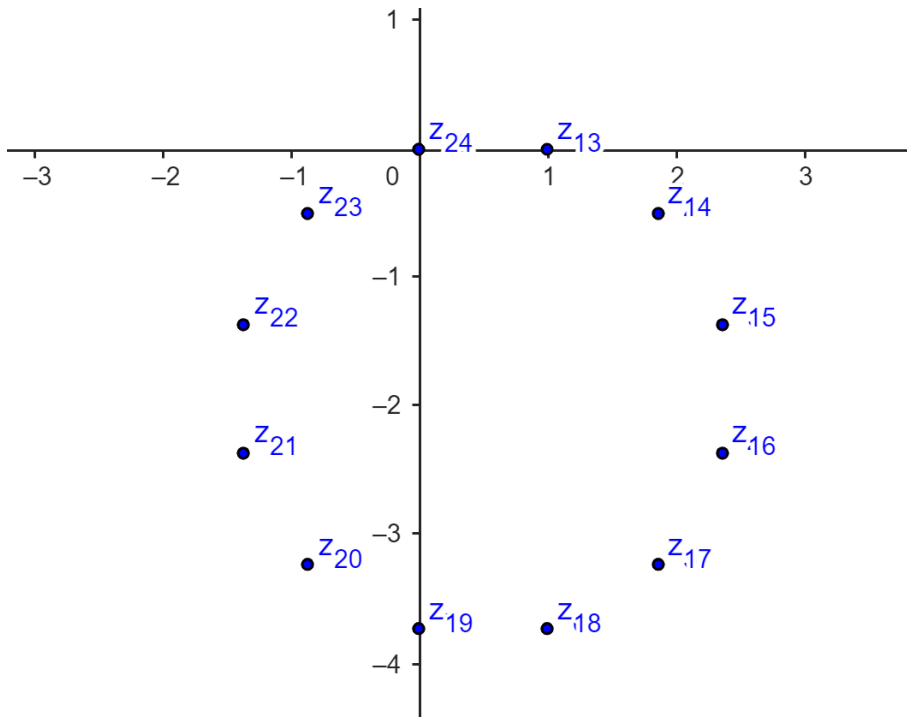
For case 1, $k = 1$, $\alpha = \frac{\pi}{12}$



For case 2, $k = 5$, $\alpha = \frac{5\pi}{12}$



For case 3, $k = 2$, $\alpha = \frac{\pi}{6}$



iv. For the test to be successful, the following condition must be met:

$$|1 + e^{-i\alpha} + e^{-i2\alpha} + e^{-i3\alpha} + \dots + e^{-i23\alpha}| \leq 1$$

$$\begin{aligned} 1 + e^{-i\alpha} + e^{-i2\alpha} + e^{-i3\alpha} + \dots + e^{-i23\alpha} &= \frac{1 - e^{-24i\alpha}}{1 - e^{-i\alpha}} \\ &= \frac{e^{-12i\alpha}(e^{12i\alpha} - e^{-12i\alpha})}{e^{-i\frac{\alpha}{2}}(e^{i\frac{\alpha}{2}} - e^{-i\frac{\alpha}{2}})} \\ &= e^{-\frac{23}{2}i\alpha} \times \frac{2i \sin(12\alpha)}{2i \sin(\frac{\alpha}{2})} \end{aligned}$$

1 mark

$$\therefore \left| e^{-\frac{23}{2}i\alpha} \times \frac{2i \sin(12\alpha)}{2i \sin(\frac{\alpha}{2})} \right| \leq 1$$

1 mark

$$\left| e^{-\frac{23}{2}i\alpha} \right| \times \left| \frac{2i \sin(12\alpha)}{2i \sin(\frac{\alpha}{2})} \right| \leq 1$$

Since $\left|e^{-\frac{23}{2}i\alpha}\right| = 1$

$$\therefore \left| \frac{2i \sin(12\alpha)}{2i \sin(\frac{\alpha}{2})} \right| \leq 1$$

1 mark

$$|2i \sin(12\alpha)| \leq \left| 2i \sin(\frac{\alpha}{2}) \right|$$

That is:

$$|\sin(12\alpha)| \leq \left| \sin(\frac{\alpha}{2}) \right|$$

■

1 mark